

SOC Alert Triage & Phishing Investigation Lab

Platform: TryHackMe - SOC Immersive Simulation

SIEM: Splunk Enterprise 8.2.6

Analyst: Abhishek Nangare

Environment: Simulated SOC environment

Primary Threat Category: Phishing / Social Engineering

Investigation Date: 2 July 2026

1. Executive Summary

This project documents a hands-on SOC alert-triage and phishing-investigation exercise performed in a simulated enterprise environment.

The investigation involved reviewing inbound email alerts, examining email telemetry in Splunk, correlating suspicious email activity with firewall events, investigating IP reputation using AbuseIPDB, extracting attack indicators, classifying alerts as true or false positives, assessing escalation requirements, and documenting findings through incident case reports.

Three principal alerts were investigated during the exercise:

- **Alert 8814:** Inbound email containing a suspicious external link - classified as a **false positive**
- **Alert 8815:** Package-delivery-themed phishing email - classified as a **true positive**
- **Third phishing alert:** Microsoft-themed unusual sign-in email - classified as a **true positive**

The simulation was successfully completed. The final simulator result states that all true-positive alerts were identified, while the recorded true-positive rate was **60%**. The simulator also noted that MTTR and dwell time were longer than average, with the phishing investigation taking notably longer to close.

2. Objectives

The objectives of the investigation were to:

1. Triage SOC alerts and determine whether they represented genuine security incidents.
2. Analyze email-related telemetry in Splunk.
3. Correlate email events with firewall/network telemetry.
4. Identify potentially relevant indicators such as sender addresses, URLs and IP addresses.
5. Use external reputation intelligence to supplement investigation findings.
6. Distinguish true positives from false positives.
7. Determine whether an alert warranted escalation.
8. Document the investigation and recommended response actions.

3. Tools and Technologies

Splunk Enterprise 8.2.6

Splunk was used as the primary SIEM during the investigation. Searches were performed against email and firewall telemetry to locate events associated with users, senders, recipients, URLs, source IPs and destination IPs.

AbuseIPDB

AbuseIPDB was used as an external IP-reputation source during investigation of network indicators.

The results were treated as supporting evidence rather than as the sole basis for determining whether an event was malicious.

SOC Immersive Simulation

The exercise provided an alert queue, SIEM data, network telemetry and an incident-reporting workflow that allowed the investigation process to be documented from alert intake through classification and case reporting.

4. Investigation Methodology

The investigation followed a SOC-style workflow:

Alert intake → Initial triage → Email analysis → SIEM investigation → Network correlation → Threat-intelligence lookup → Classification → Escalation assessment → Case documentation

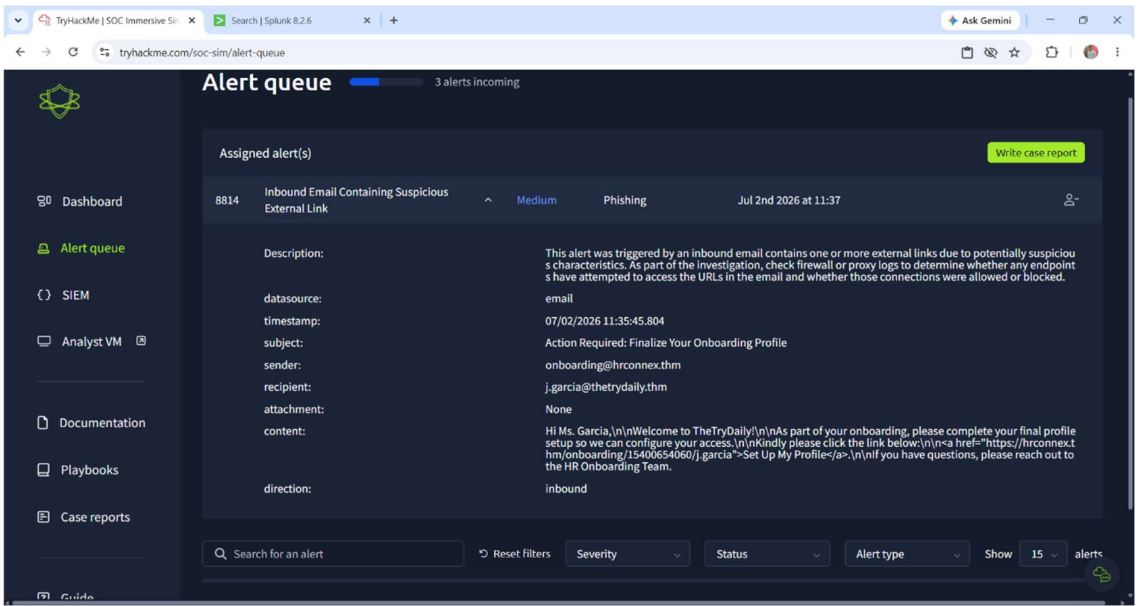
The process was repeated for each alert.

5. Incident 1 - Inbound Email Containing Suspicious External Link

Alert Information

Alert ID: 8814
Alert Type: Phishing
Severity: Medium
Data Source: Email
Alert Time: 2 July 2026 at approximately 11:37
Event Timestamp: 07/02/2026 11:35:45.804
Subject: Action Required: Finalize Your Onboarding Profile
Sender: onboarding@hrconnex.thm
Recipient: j.garcia@thetrydaily.thm
Attachment: None
Direction: Inbound

The alert was generated because the inbound message contained an external URL that triggered the suspicious-link detection rule.



Initial Analysis

The email instructed the recipient to complete an onboarding profile and contained a link to an `hrconnex.thm` onboarding resource.

Rather than immediately treating the alert as malicious, related email telemetry was investigated in Splunk.

The search results showed additional internal communication involving the same recipient, including messages related to hiring/interview scheduling and internal event coordination. The onboarding email was therefore reviewed in the broader context of the user's normal organizational activity.

SIEM Investigation

The Splunk investigation located multiple email events associated with `j.garcia@thetrydaily.thm`.

Examples of related activity included:

- internal hiring/interview communication
- internal event-related communication
- the onboarding notification itself

This surrounding context supported the interpretation that the onboarding message was part of a legitimate internal process rather than an isolated malicious communication.

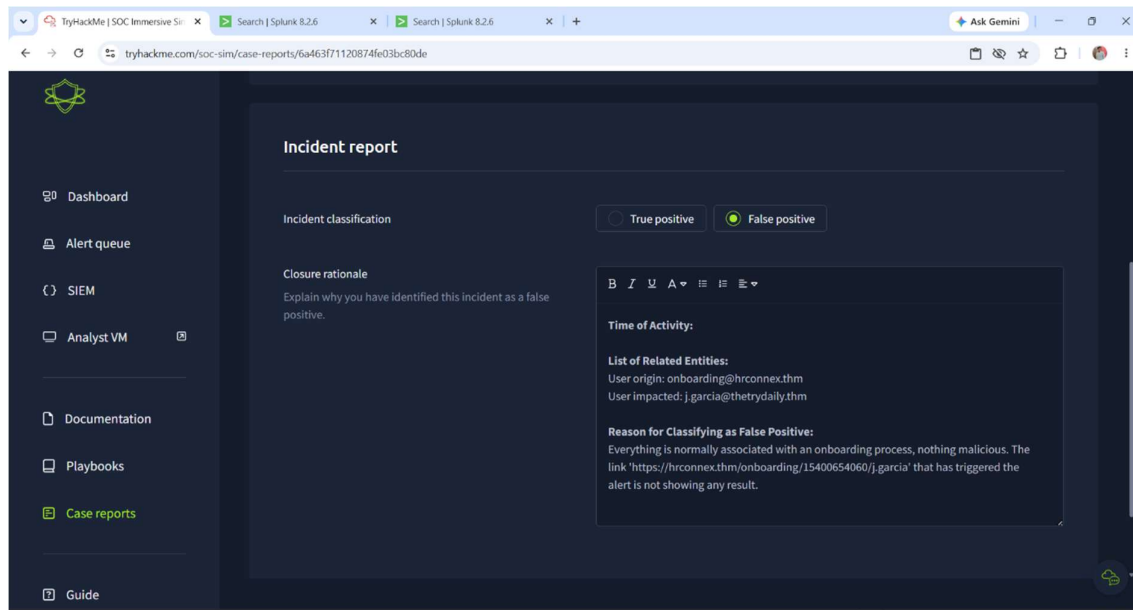
Classification

Classification: False Positive

The simulator case report states that the activity was normally associated with the onboarding process and that no malicious behavior was identified in the investigation. The report also notes that the link associated with the alert was not producing a suspicious result in the observed context.

Outcome

The alert was closed as a **false positive**.



Analyst Takeaway

This investigation demonstrates the importance of **contextual alert validation**. A suspicious-link alert should not automatically be treated as a confirmed phishing incident without examining the surrounding user and communication context.

6. Incident 2 - Package Delivery Phishing Email

Alert Information

Alert ID: 8815
Alert Type: Phishing
Severity: Medium
Data Source: Email
Alert Time: 2 July 2026 at approximately 11:40
Event Timestamp: 07/02/2026 11:38:58.804
Subject: Your Amazon Package Couldn't Be Delivered - Action Required
Sender: urgents@amazon.biz
Recipient: h.harris@thetrydaily.thm
Attachment: None
Direction: Inbound

The email used a package-delivery theme and instructed the recipient to confirm shipping information through a shortened URL.

The screenshot shows a web browser window with multiple tabs. The active tab is 'tryhackme.com/soc-sim/alert-queue'. The browser's address bar shows the URL. The page content is a dark-themed SIEM interface. On the left is a sidebar with navigation links: Dashboard, Alert queue (highlighted), SIEM, Analyst VM, Documentation, Playbooks, and Case reports. The main area is titled 'Alert queue' with a sub-header '0 alerts incoming'. Below this is a table of 'Assigned alert(s)'. The table has columns for ID, Description, Severity, Type, and Time. One alert is listed: ID 8815, Description 'Inbound Email Containing Suspicious External Link', Severity 'Medium', Type 'Phishing', and Time 'Jul 2nd 2026 at 11:40'. To the right of the table is a 'Write case report' button. Below the table, the details of the selected alert are shown in a two-column format. The left column lists fields: Description, datasource, timestamp, subject, sender, recipient, attachment, content, and direction. The right column provides the corresponding values. The 'content' field contains a phishing email body text. At the bottom of the interface is a search bar and filter controls.

ID	Description	Severity	Type	Time
8815	Inbound Email Containing Suspicious External Link	Medium	Phishing	Jul 2nd 2026 at 11:40

Alert Details:

- Description:** This alert was triggered by an inbound email contains one or more external links due to potentially suspicious characteristics. As part of the investigation, check firewall or proxy logs to determine whether any endpoints have attempted to access the URLs in the email and whether those connections were allowed or blocked.
- datasource:** email
- timestamp:** 07/02/2026 11:38:58.804
- subject:** Your Amazon Package Couldn't Be Delivered - Action Required
- sender:** urgents@amazon.biz
- recipient:** h.harris@thetrydaily.thm
- attachment:** None
- content:** Dear Customer,\n\nWe were unable to deliver your package due to an incomplete address.\n\nPlease confirm your shipping information by clicking the link below:\n\nhttp://bit.ly/3sHkX3da12340\n\nIf we don't hear from you within 48 hours, your package will be returned to sender.\n\nThank you,\n\nAmazon Delivery
- direction:** inbound

6.1 Email Analysis

The sender address used the domain `amazon.biz`, while the message presented itself as an Amazon delivery notification.

The email contained a shortened Bitly URL:

`http://bit.ly/3sHkX3da12340`

The use of a shortened URL makes the final destination less immediately visible to the recipient and therefore warranted further investigation.

6.2 Splunk Investigation

A Splunk search was performed against the sender and recipient information to locate the associated email event.

The relevant email record confirmed:

- **sender:** `urgents@amazon.biz`
- **recipient:** `h.harris@thetrydaily.thm`
- **subject:** `Your Amazon Package Couldn't Be Delivered - Action Required`
- **timestamp:** `07/02/2026 11:38:58.804`
- **attachment:** `none`
- **direction:** `inbound`

The message content contained the shortened Bitly link.

6.3 Firewall Correlation

The investigation was expanded to firewall telemetry.

A corresponding firewall event was identified at:

Timestamp: `07/02/2026 11:40:12.804`

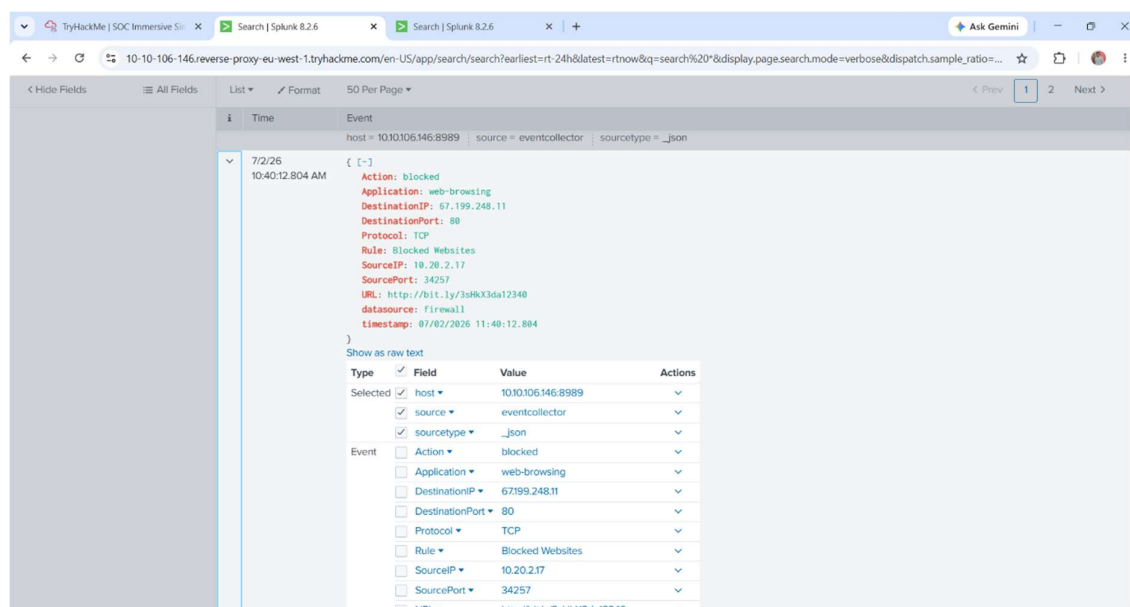
The event contained the following fields:

Field	Value
Action	blocked
Application	web-browsing
Destination IP	67.199.248.11
Destination Port	80
Protocol	TCP
Rule	Blocked Websites
Source IP	10.20.2.17

Field	Value
Source Port	34257
URL	http://bit.ly/3sHkX3da12340
Data Source	firewall

The firewall therefore recorded an attempted web connection to the Bitly URL that was **blocked by the Blocked Websites rule**.

A separate firewall event from the same source IP showed normal browsing activity to Google over HTTPS, demonstrating that the investigation was distinguishing the suspicious connection from ordinary allowed internet traffic rather than treating all network activity from the host as malicious.



6.4 IP Reputation Investigation

The destination IP 67.199.248.11 was investigated using AbuseIPDB.

At the time shown in the screenshot:

- the IP was present in the AbuseIPDB database
- it had **851 reports**
- displayed **Confidence of Abuse: 0%**
- **ISP: Bitly Inc**
- **Usage Type: Content Delivery Network**
- **Hostname: bit.ly**
- **Domain: bit.ly**

Important Analytical Observation

The AbuseIPDB result **did not establish that the IP itself was malicious** because the displayed abuse-confidence value was 0%.

The result instead helped establish that the destination belonged to Bitly infrastructure. The maliciousness assessment therefore depended on the **overall evidence**, including the phishing email, sender/domain characteristics, shortened URL and associated firewall activity.

This is an important distinction between **indicator enrichment** and **indicator confirmation**.

The screenshot shows the AbuseIPDB website interface. At the top, there's a navigation bar with links like 'Report IP', 'Bulk Checker', 'Bulk Reporter', 'Pricing', 'Integrations', 'Docs', 'IP Utilities', 'Contact', and 'More'. Below this, a green banner displays the IP address '5.188.10.0/24, or AS15169'. The main content area shows a message: '67.199.248.11 was found in our database!'. Below this, it states 'This IP was reported 851 times. Confidence of Abuse is 0%'. A green bar indicates the 0% confidence level. To the right, there's a sidebar with a 'feedback' button and a 'Shutterstock Free Trial' advertisement. The main content area lists various details about the IP: ISP (Bitly Inc), Usage Type (Content Delivery Network), ASN (AS396982), Hostname(s) (bit.ly), Domain Name (bit.ly), Country (United States of America), and City (New York City, New York). At the bottom, there are two buttons: 'REPORT 67.199.248.11' and 'WHOIS 67.199.248.11'. A footer note mentions 'IP Abuse Reports for 67.199.248.11'.

6.5 Classification

Classification: True Positive

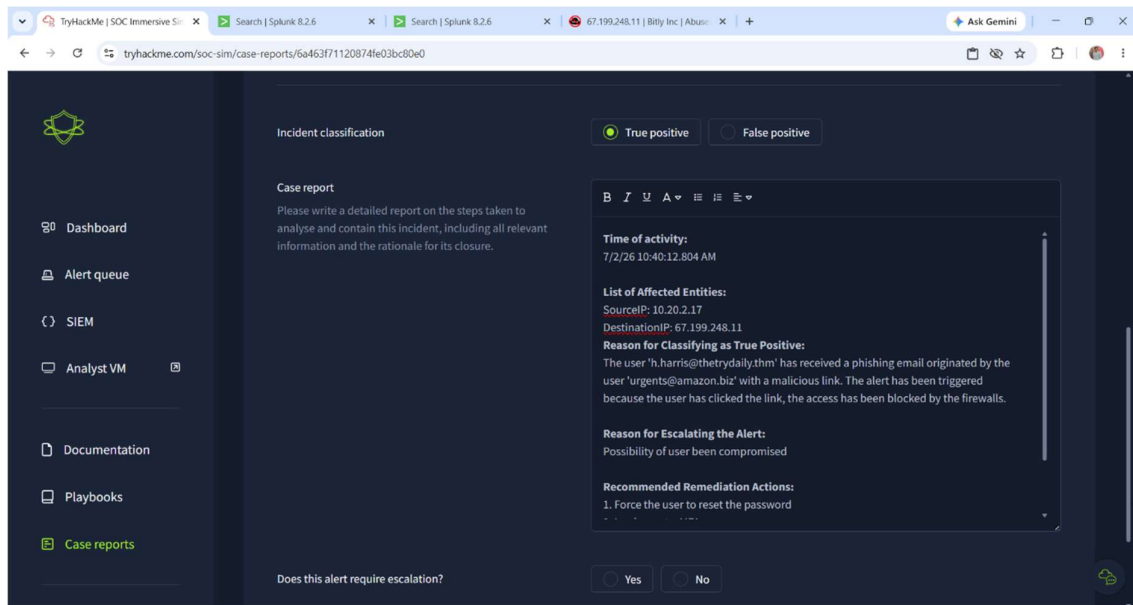
The simulator's case report identified the event as a phishing attempt and stated that the link was associated with the user activity and that the firewall blocked access. The report identified the possibility that the user's account had been compromised.

6.6 Escalation Assessment

The case report records the reason for escalation as:

Possibility of user being compromised.

This should be described as an **escalation assessment/reason**, not as proof that a real-world escalation was executed.



6.7 Recommended Remediation

The case report listed:

1. Force the user to reset the password.
2. Implement MFA.

These are **recommended remediation actions shown in the simulator case report**. The screenshots do not demonstrate that the password reset or MFA change was actually performed.

6.8 Attack Indicators

The case report identifies the malicious URL as an attack indicator. The investigation also identified:

- suspicious sender address
- suspicious sender domain
- shortened phishing URL
- destination IP associated with the URL
- affected source IP

7. Incident 3 - Microsoft-Themed Unusual Sign-In Phishing

Alert Information

Alert Type: Phishing / Suspicious Email

Data Source: Email

Event Timestamp: 07/02/2026 11:41:16.804

Subject: Unusual Sign-In Activity on Your Microsoft Account

Sender: no-reply@microsoftsupport.co

Recipient: c.allen@thetrydaily.thm

Attachment: None

Direction: Inbound

The email claimed that an unusual Microsoft account sign-in had occurred and provided a link for the recipient to review the activity.

7.1 Email Analysis

The sender used:

no-reply@microsoftsupport.co

The email link pointed to:

<https://microsoftsupport.co/login>

The message also included a claimed sign-in location of:

Lagos, Nigeria

and the IP:

102.89.222.143

The event therefore contained multiple phishing indicators requiring investigation.

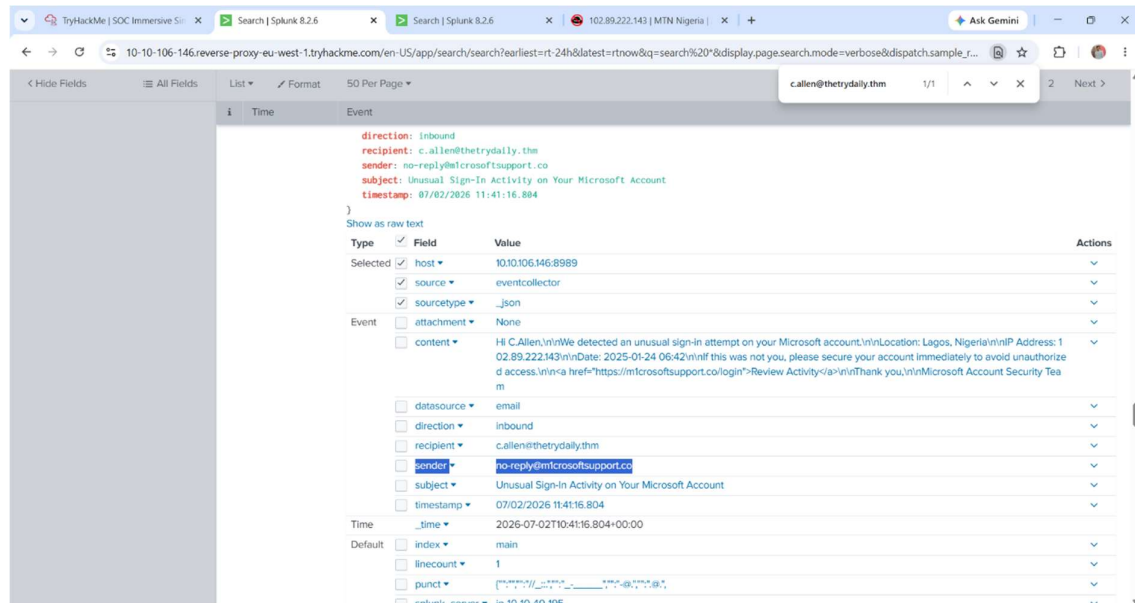
7.2 Splunk Investigation

A Splunk search was performed against the sender address:

no-reply@microsoftsupport.co

The resulting event confirmed the inbound message and its associated fields, including sender, recipient, subject and timestamp.

The investigation also showed the email content containing the Microsoft-themed sign-in warning and the external login link.



7.3 IP Reputation Investigation

The IP 102.89.222.143 was checked using AbuseIPDB.

The screenshot showed:

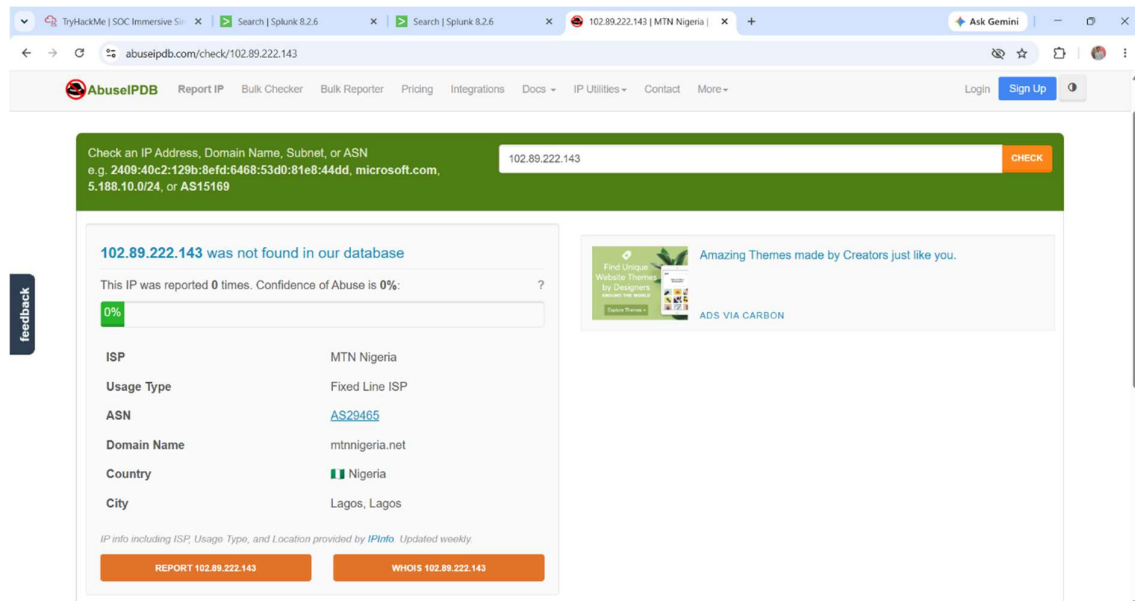
- **IP not found in the database**
- **0 reports**
- **Confidence of Abuse: 0%**
- **ISP: MTN Nigeria**
- **Usage Type: Fixed Line ISP**
- **Country: Nigeria**
- **City: Lagos**

Analytical Interpretation

This result does **not** prove that the IP was legitimate, nor does it prove that the IP was fabricated.

It simply means that this particular reputation source did not provide positive abuse evidence for the IP at the time of the lookup.

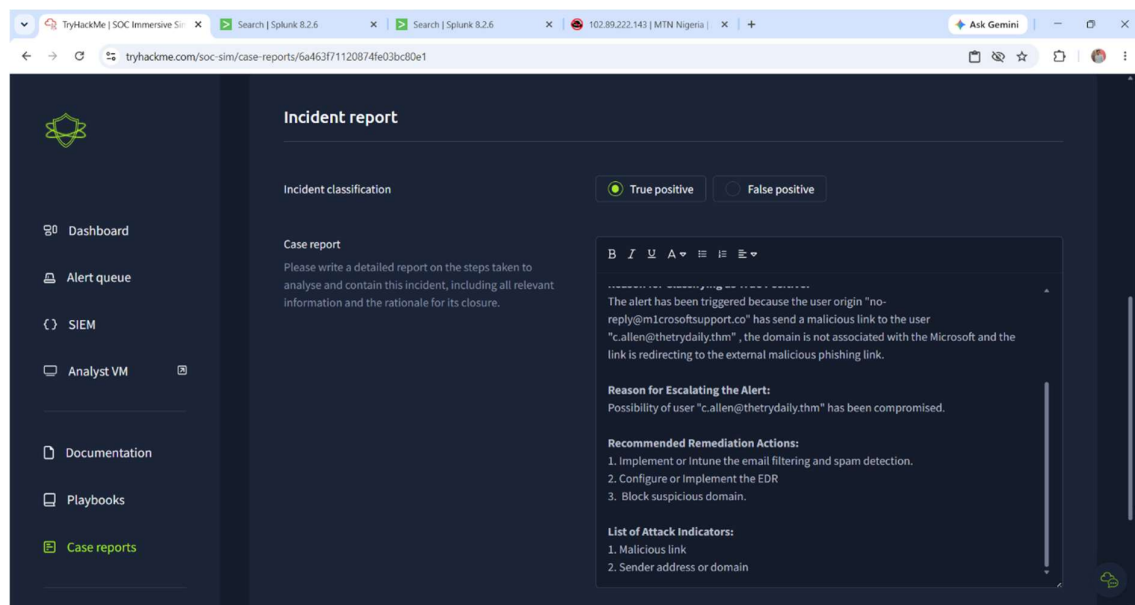
The phishing assessment therefore relied primarily on the email's other characteristics, particularly the sender/domain and suspicious login URL.



7.4 Classification

Classification: True Positive

The case report classified the alert as a true positive because the sender had sent a malicious link to the affected user, the domain was not associated with Microsoft, and the link redirected toward an external phishing destination.



7.5 Escalation Assessment

The case report states:

Possibility of user `c.allen@thetrydaily.thm` being compromised.

Again, this should be presented as the simulator's **reason for escalation**, not as evidence of an actual production compromise.

7.6 Recommended Remediation

The case report recommends:

1. Implement or tune email filtering and spam detection.
2. Configure or implement EDR.
3. Block the suspicious domain.

The screenshots do not show these controls being actually deployed.

7.7 Attack Indicators

The investigation identified:

- suspicious sender address
- suspicious sender domain
- phishing login URL
- IP referenced in the email
- affected recipient account

8. Investigation Comparison

Alert	Threat Theme	Classification	Main Evidence
8814	Onboarding email	False Positive	Supporting internal communication and normal onboarding context
8815	Package-delivery phishing	True Positive	Suspicious sender/domain, shortened URL, related firewall event and blocked connection
Third phishing alert	Microsoft-themed sign-in phishing	True Positive	Suspicious sender/domain and external login URL

The three investigations demonstrate why SOC alert triage requires **context and correlation rather than alert-name-based decisions**.

9. SOC Investigation Workflow Demonstrated

This exercise provided hands-on practice with the following activities.

Alert Triage

Reviewed incoming alerts and determined whether additional investigation was required.

Email Analysis

Examined:

- sender
- recipient
- subject
- message content
- embedded URLs
- attachment status
- direction of email traffic

SIEM Searching

Used Splunk to search by email attributes such as:

- sender
- recipient
- source IP
- destination IP

Log Correlation

Correlated email telemetry with firewall telemetry to establish whether suspicious links were associated with network activity.

Network Analysis

Reviewed:

- source IP
- destination IP
- source/destination ports
- protocol
- URL
- firewall action
- security rule
- application

Threat Intelligence

Used AbuseIPDB to enrich IP indicators and interpret their reputation results.

Alert Classification

Distinguished between:

- legitimate activity triggering a security rule
- genuine phishing activity requiring further response

Incident Documentation

Created case reports documenting:

- time of activity
- affected entities
- classification rationale
- escalation reason
- remediation recommendations
- attack indicators

10. Key Findings

Finding 1 - False positives require contextual validation

The onboarding alert demonstrates that a suspicious external link can occur within an otherwise legitimate business process. Reviewing surrounding communication in the SIEM helped establish the context before closing the alert.

Finding 2 - Correlation increases confidence

The package-delivery phishing investigation became more significant when the email evidence was correlated with firewall telemetry showing a related network connection that was blocked.

Finding 3 - Threat intelligence is supporting evidence

The AbuseIPDB results demonstrate that an IP reputation lookup should not automatically determine the final classification. The Bitly-related IP showed 0% abuse confidence despite being present in the database, while the Microsoft-themed phishing IP had no AbuseIPDB reports.

Finding 4 - A possible compromise should influence escalation

Both true-positive case reports recorded the possibility of user compromise as part of the escalation assessment.

11. Metrics and Final Outcome

The final simulator result reported:

Scenario Status: Victory! Security breach prevented!

True-Positive Rate: 60%

The simulator stated that all true-positive alerts were identified, while also reporting that MTTR and dwell time were longer than average and that a phishing alert took notably longer to close.

Important distinction

The **60% true-positive rate should not be changed to 100%** in your portfolio.

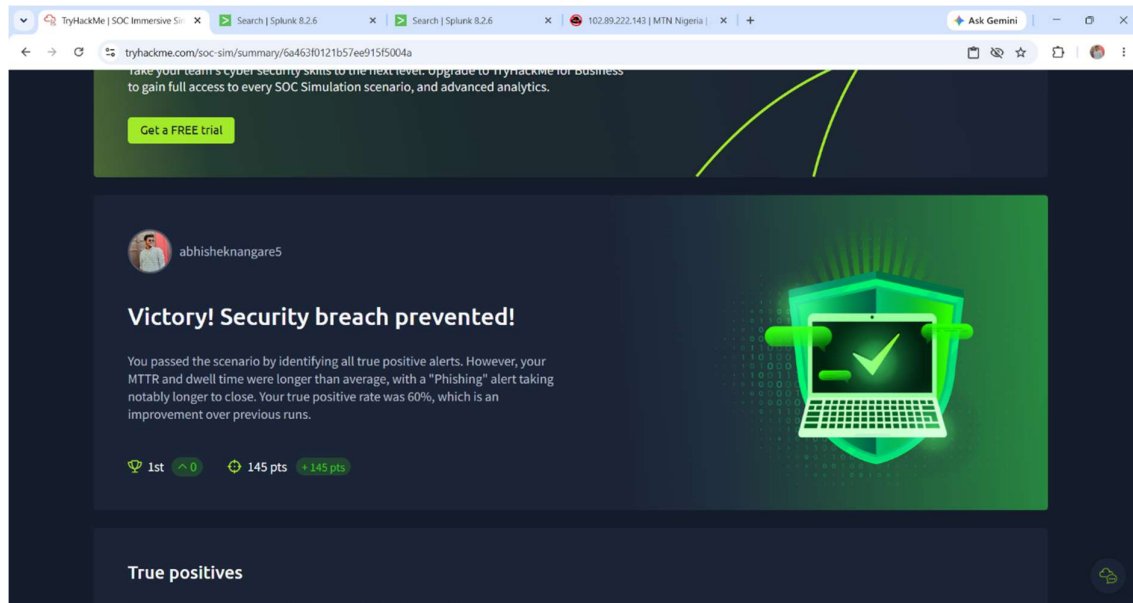
The statements:

"All true-positive alerts were identified"

and

"True-positive rate was 60%"

are not contradictory. The first describes whether the actual true-positive alerts were successfully recognized; the second is the simulator's broader performance metric.



12. Limitations

This investigation was conducted in a **simulated SOC environment**, so the findings should not be presented as a real corporate security incident.

The screenshots demonstrate investigation and documentation activity but do not prove that the recommended remediation actions were performed in an actual environment.

Similarly, an IP reputation result from AbuseIPDB is only one source of evidence and should be interpreted together with email, network and endpoint telemetry.

The screenshots also show searches using a **24-hour investigation window**, meaning the investigation evidence shown represents the observed search scope rather than proof that no activity existed outside that period.

13. Skills Demonstrated

SOC Alert Triage
Phishing Investigation
Email Security Analysis
Splunk SIEM
SPL Searching
Log Analysis
Firewall Log Analysis
Event Correlation
IOC Extraction
IP Reputation Analysis
Threat Intelligence

True Positive / False Positive Classification
Incident Documentation
Escalation Assessment
Incident Response Fundamentals

14. Conclusion

This lab provided practical exposure to a SOC analyst workflow beginning with alert intake and continuing through SIEM investigation, network correlation, threat-intelligence enrichment, alert classification and incident documentation.

The investigations demonstrated three important analyst behaviors:

1. **Do not assume every alert is malicious.**
2. **Correlate multiple telemetry sources before reaching a conclusion.**
3. **Use threat intelligence as supporting evidence rather than relying on a single reputation result.**

The exercise ultimately resulted in successful identification of the simulated true-positive alerts and completion of the scenario.